

DARES Electronics: Autonomy and Perception

Week 9 and Mid-semester Break

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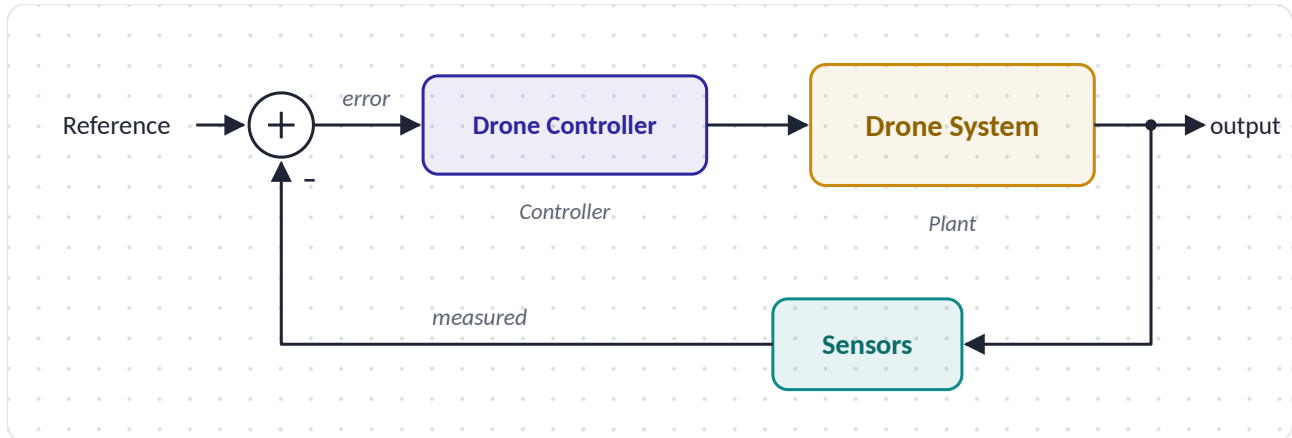


Figure 1: High level block diagram

Our choices:

- What do we measure: all 6 DOF (pitch, roll, yaw, x, y, z). Can we also get velocities? Can we get accelerations?
- What do we control? Position control? Velocity control? Ideally we work towards full state control, where we do both at once.
- What does our controller look like? Starting point: PID. Ideally we work towards LQR (optimal controller).

General Process

- We will need to create a **state space model** of the drone, relating inputs (from the motors) to outputs (pitch, roll, yaw etc).
- This model does not have to be perfect, but should be reasonably accurate.
- This will rely on theoretical numbers from propulsion, alongside real testing.
- Until we reach the point of having a physical drone, we can still create the framework for our controller, and improve it as we get more and more refined information.

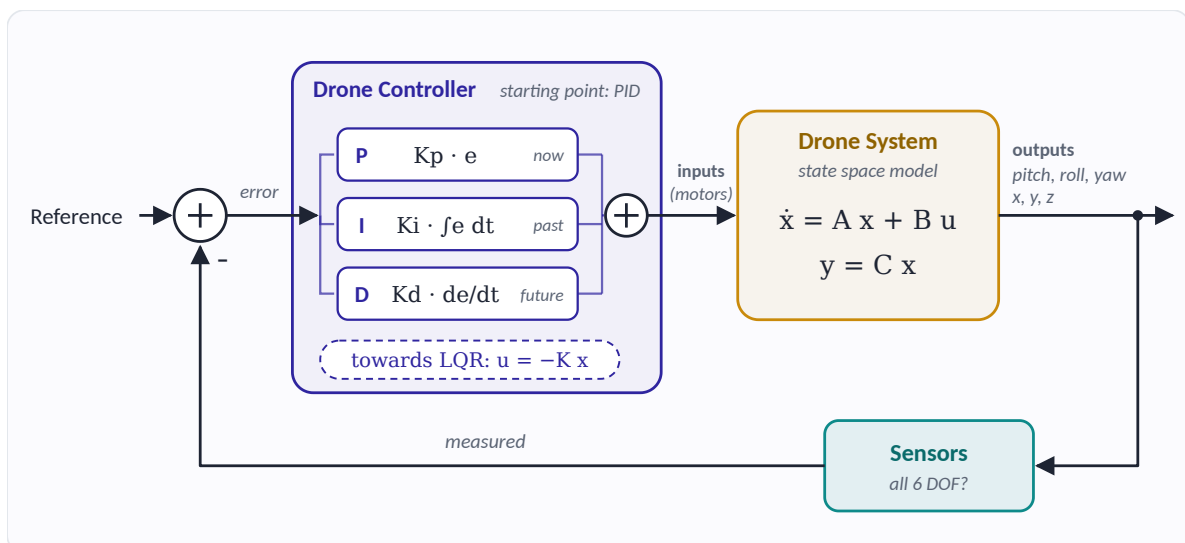


Figure 2: Figure 1, opened up



Software

- Theoretical controller design will be done in MATLAB.
- Final implementation will likely be done through a microcontroller in something like C/C++.

To-Do Over Midsem

- A lot of the topics we will be dealing with don't get covered until early masters. However, there are tons of resources, and a lot is very intuitive to get the big picture of.
- By the end of midsem, make sure you have watched these videos to get up to speed:

- 1 **Closed Loop Control** (Systems Modelling and Analysis, Control Systems) - Overview of what CL control is (reading sensors and using that to correct position)
- 2 **Time/Frequency Domain** (SMA, Signals and Systems, Analog and Digital Electronics Concepts, Electrical Network Analysis and Design) - Simplifies complicated integrals into easy multiplication, allows us to design controllers
- 3 **Transfer Functions & Laplace** (Engineering Maths (kind of), SMA, SS, ADEC, ENAD) - lets us move into frequency domain and see how inputs affect outputs
- 4 **System Stability and Control** (Control Systems) - makes sure that when we design a controller it does what we want instead of growing out of control
- 5 **PID Control** (Control Systems) - faster than simple control, makes sure we have no error long term
- 6 **PID** Examples - intuitive examples related to PID

Further (optional) research

- If you are interested in some of the analysis techniques we could use, have a look at Brian Douglas' videos on Bode Plots, and Gain/Phase Margins.
- If you are interested in learning about some of the **Advanced Control** techniques for optimal controllers, have a look at [this video](#)